CHAPTER 1

The Fundamentals of Managerial Economics

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Learning Objectives

1. Summarize how goals, constraints, incentives, and market rivalry affect economic decisions.
2. Distinguish economic versus accounting profits and costs.
3. Explain the role of profits in a market economy.
4. Apply the five forces framework to analyze the sustainability of an industry's profits.
5. Apply present value analysis to make decisions and value assets.
6. Apply marginal analysis to determine the optimal level of a managerial control variable.
7. Identify and apply six principles of effective managerial decision making.

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The Manager

- A person who directs resources to achieve a stated goal.
  - Directs the efforts of others.
  - Purchases inputs used in the production of the firm's output.
  - Directs the product price or quality decisions.

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Economics

- The science of *making decisions* in the presence of *scarce resources*.
  - *Resources* are anything used to produce a good or service, or achieve a goal.
  - *Decisions* are important because scarcity implies trade-offs.

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Managerial Economics Defined

- The study of how to direct scarce resources in the way that most efficiently achieves a managerial goal.
  - Should a firm purchase components – like disk drives and chips – from other manufacturers or produce them within the firm?
  - Should the firm specialize in making one type of computer or produce several different types?
  - How many computers should the firm produce, and at what price should you sell them?

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Economics of Effective Management

- Basic principles comprising effective management:
  - Identify goals and constraints
  - Recognize the nature and importance of profits
  - Understand incentives
  - Understand markets
  - Recognize the time value of money
  - Use marginal analysis

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Identify Goals and Constraints

- Well-defined goals
- Firm's overall goal is to maximize profits
- Constraints make it difficult to achieve goals
  - Available technology
  - Prices of inputs used in production

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Recognize the Nature and Importance of Profits

- **Accounting profit**
  - Total amount of money taken in from sales (total revenue) minus the dollar cost of producing goods or services.
- **Economic profit**
  - The difference between total revenue and cost *opportunity* cost.
  - **Opportunity cost**
    - The explicit cost of a resource plus the implicit cost of giving up its best alternative.

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Recognize the Nature and Importance of Profits

- **The role of profits**
  - Profits are a signal to resource holders where resources are most highly valued by society.

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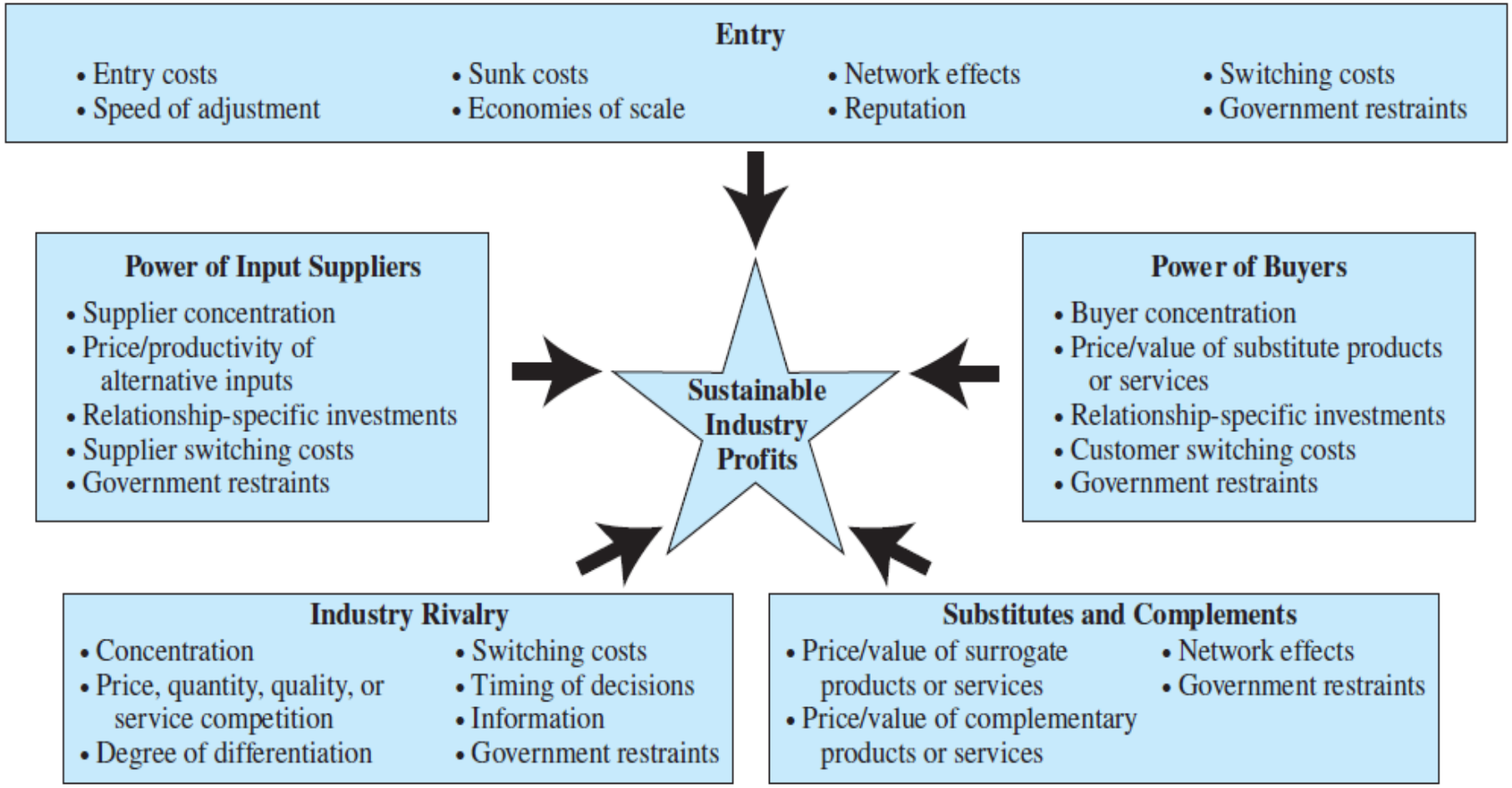
Five Forces and Industry Profitability



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Understand Incentives

- Changes in profits provide an *incentive* to how resource holders use their resources.
- Within a firm, incentives impact how resources are used and how hard workers work.
  - One role of a manager is to construct incentives to induce maximal effort from employees.

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Understand Markets

- Two sides to every market transaction: buyer and seller
- Bargaining position of consumers and producers is limited by three rivalries in economic transactions:
  - Consumer-producer rivalry
  - Consumer-consumer rivalry
  - Producer-producer rivalry
- Government and the market

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Recognize the Time Value of Money

- Often a gap exists between the time when costs are borne and benefits received.
  - Managers can use *present value analysis* to properly account for the timing of receipts and expenditures.

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Present Value Analysis 1

- Present value of a *single* future value
  - The amount that would have to be invested today at the prevailing interest rate to generate the given future value:

$$PV = \frac{FV}{(1+i)^n}$$

  - Present value reflects the difference between the *future value* and the *opportunity cost of waiting*:

$$PV = FV - OCW$$

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Present Value Analysis II

- Present value of a *stream of* future values

$$PV = \frac{FV_1}{(1+i)^1} + \frac{FV_2}{(1+i)^2} + \cdots + \frac{FV_n}{(1+i)^n}$$

or,

$$PV = \sum_{t=1}^{n} \frac{FV_t}{(1+i)^t}$$

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The Time Value of Money in Action

- Consider a project that returns the following income stream:
  - Year 1, $10,000; Year 2, $50,000; and Year 3, $100,000.
  - At an annual interest rate of 3 percent, what is the present value of this income stream?

$$PV = \frac{\$10{,}000}{(1+0.03)^1} + \frac{\$50{,}000}{(1+0.03)^2} + \frac{\$100{,}000}{(1+0.03)^3}$$
$$= \$148{,}352.70$$

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Net Present Value

- The present value of the *income stream* generated by a project minus the current cost of the project:

$$NPV = \frac{FV_1}{(1+i)^1} + \frac{FV_2}{(1+i)^2} + \cdots + \frac{FV_n}{(1+i)^n} - C_0$$

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Present Value of Indefinitely Lived Assets

- Present value of decisions that indefinitely generate cash flows:

$$PV_{Asset} = CF_0 + \frac{CF_1}{(1+i)^1} + \frac{CF_2}{(1+i)^2} + \frac{CF_3}{(1+i)^3} + \cdots$$

- Present value of this perpetual income stream when the same cash flow is generated ($CF_1 = CF_2 = \cdots = CF$):

$$PV_{Perpetuity} = \frac{CF}{i}$$

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Present Value and Profit Maximization

- Profit maximization
  - Maximizing profits means maximizing the value of the firm, which is the present value of current and future profits.

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Present Value and Estimating Values of Firms I

- The value of a firm with *current profits* π_0, with *no dividends paid out* and *expected, constant profit growth rate* of g (assuming $g < i$) is:

$$PV_{Firm} = \pi_0 + \frac{\pi_0(1+g)}{(1+i)^1} + \frac{\pi_0(1+g)^2}{(1+i)^2} + \frac{\pi_0(1+g)^3}{(1+i)^3} + \cdots$$

$$= \pi_0 \left(\frac{1+i}{i-g}\right)$$

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Present Value and Estimating Values of Firms II

- When dividends are immediately paid out of current profits, the present value of the firm is (at ex-dividend date):

$$PV_{Firm}{}^{Ex-div} = PV_{Firm} - \pi_0$$
$$= \pi_0 \left(\frac{1+g}{i-g} \right)$$

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Short-Term versus Long-Term Profits

- Short-term and long-term profits
  - If the growth rate in profits is less than the interest rate and both are constant, maximizing current (short-term) profits is the same as maximizing long-term profits.

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Use Marginal Analysis

- Given a control variable, Q, of a managerial objective, denote the
  - total benefit as $B(Q)$.
  - total cost as $C(Q)$.
- Manager's objective is to maximize net benefits:

$$N(Q) = B(Q) - C(Q)$$

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Use Marginal Analysis

- How can the manager maximize net benefits?
- Use marginal analysis
  - **Marginal benefit:** $MB(Q)$
    - The change in total benefits arising from a change in the managerial control variable, Q.
  - **Marginal cost:** $MC(Q)$
    - The change in the total costs arising from a change in the managerial control variable, Q.
  - Marginal net benefits: $MNB(Q)$

$$MNB(Q) = MB(Q) - MC(Q)$$

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Use Marginal Analysis

- Marginal principle
  - To maximize net benefits, the manager should increase the managerial control variable up to the point where marginal benefits equal marginal costs. This level of the managerial control variable corresponds to the level at which marginal net benefits are zero; nothing more can be gained by further changes in that variable.

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Marginal Analysis In Action

- It is estimated that the benefit and cost structure of a firm is:

$$B(Q) = 250Q - 4Q^2$$
$$C(Q) = Q^2$$

- Find the $MB(Q)$ and $MC(Q)$ functions.

$$MB(Q) = 250 - 8Q$$
$$MC(Q) = 2Q$$

- What value of Q makes $NMB(Q)$ zero?

$$250 - 8Q = 2Q \Rightarrow Q = 25$$

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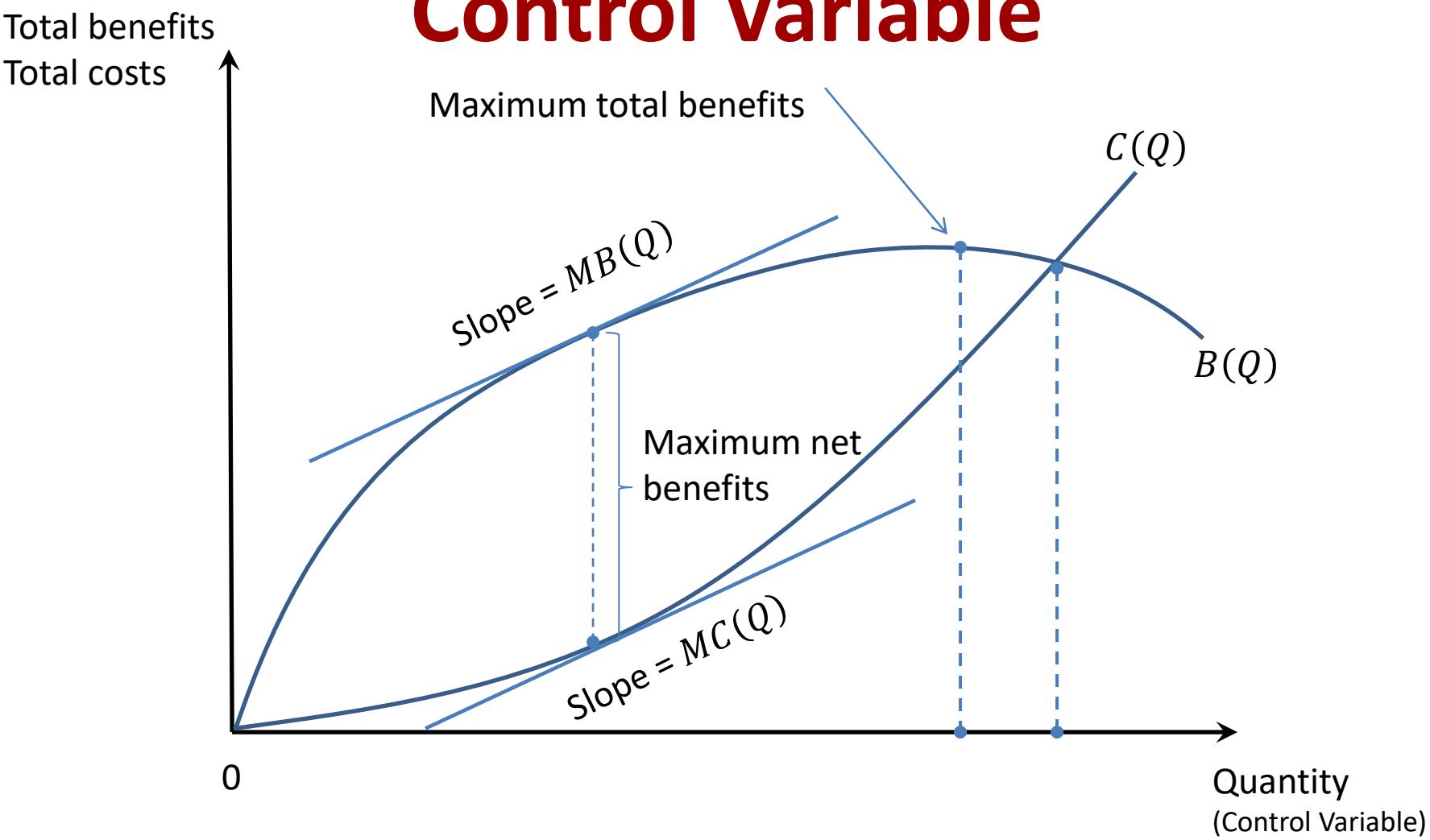
Determining the Optimal Level of a Control Variable



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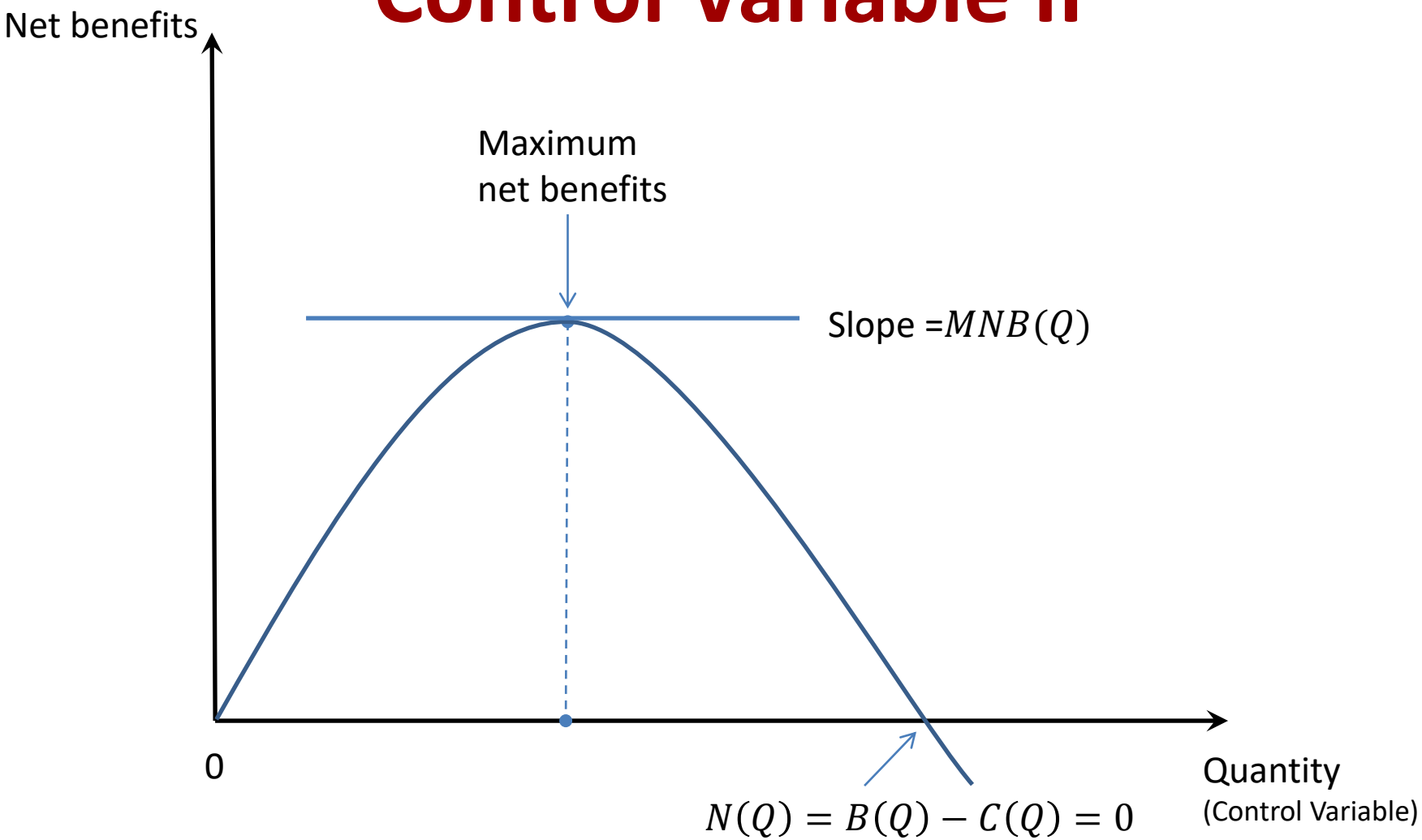
Determining the Optimal Level of a Control Variable II



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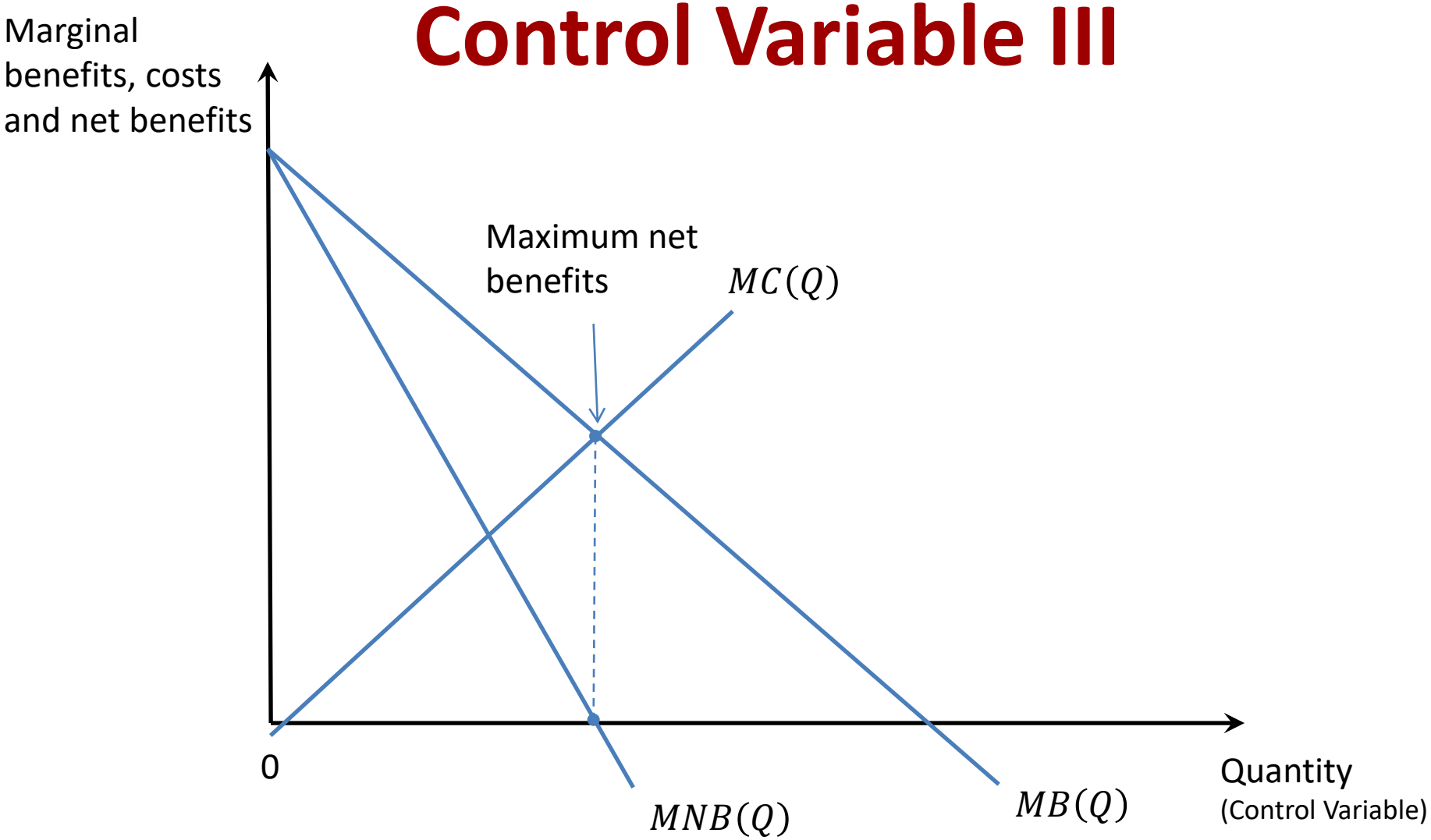
Determining the Optimal Level of a Control Variable III



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Marginal Value Curves Are the Slopes of Total Value Curves

- When the control variable is infinitely divisible, the slope of a total value curve at a given point is the marginal value at that point.
- The slope of the total benefit curve at a given *Q* is the marginal benefit of that level of *Q*.
- The slope of the total cost curve at a given *Q* is the marginal cost of that level of *Q*.
- The slope of the net benefit curve at given *Q* is the marginal net benefit of that level of *Q*.

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Marginal Value Curves Are the Slopes of Total Value Curves

- **A calculus alternative**
  - Slope of a continuous function is the derivative /marginal value of that function:

$$MB = \frac{dB(Q)}{dQ}$$

$$MC = \frac{dC(Q)}{dQ}$$

$$MNB = \frac{dN(Q)}{dQ}$$

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Incremental Decisions

- **Incremental revenues**
  - The additional revenues that stem from a yes-or-no decision.
- **Incremental costs**
  - The additional costs that stem from a yes-or-no decision.
- "Thumbs up" decision
  - $MB > MC$.
- "Thumbs down" decision
  - $MB < MC$.

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Learning Managerial Economics

- Practice, practice, practice …
- Learn terminology
  - Break down complex issues into manageable components.
  - Helps economics practitioners communicate efficiently.

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